

2/9/26

CCNA FUN TEST

① Types of medium:

- 1) Electrical signals → copper cable
- 2) Radio frequency → wireless
- 3) Light signals → Fibre

② Types of network communication:

- 1) Unicast → One to one
- 2) Broadcast → One to group
- 3) multicast → One to all.

③

PC	SWITCH
1 - T+	1 +R
2 - T-	2 -R
3 - R+	3 +T
4 +V	4 +V
5 -V	5 -V
6 R-	6 -T
7 x	7 x
8 x	8 x

④ Colour code of T568 B

white orange

Orange

white Green

Blue

White Blue

Green

white Brown

Brown.

5) PC - PC

Same functional device → cross cable.

6) Router - Switch

Different function device → straight cable.

7) Auto-MDIX

Even if we ^{have} 2 same functional or cross-functional device we can give straight or cross cable using Auto MDIX.

8) Function of wired & wireless NIC.

Wired NIC : UTP cable.

- * Physical connection

- * Data transmission.

Wireless NIC : wifi.

- * Radio frequency connection.

9) POE → Power over Ethernet

Transferring power via UTP cable.

Eg: CCTV.

10) POE switch

POE injector.

If we have to connect many devices then we can use switch (48 port).

If we have single device, then POE switch can be connected to non POE camera using injector.

11) IP PARAMETER:

181.0.20.0.

N/W IP: 181.0.0.0

First IP: 181.0.0.1

Last IP: 181.0.255.254

Broadcast IP: 181.0.255.255.

Subnet mask: 255.255.0.0.

Prefix: N/W bits $\rightarrow$ 16

Previous IP: 181.0.19.255.

Next IP: 181.0.20.1

12) Total no. of IP: class-A: 16777214

class-B: 65534

class-C: 254.

13) Why subnet mask?

Subnet mask is used to split the IP address into network & host:
specific device.

14) RDP $\rightarrow$ Remote Desktop Protocol.

We can control a computer from another device in same n/w using RDP.

15) Netbios $\rightarrow$ This protocol is used to resolve name into IP address.

16) Explain OSI layers : ping 70.10 to 70.20

ping 70.20

Application layer: Echo data is generated.

Presentation layer: Format encapsulation

Session layer: Time encapsulation

Transport layer: Port number encapsulation

N/w layer: Packet encapsulation → SIP
DIP

Data-Link Layer: Frame encapsulation → S MAC
D MAC

ARP req: →

← ARP Reply

Physical layer: Data is given out
in physical layer (wire)

17) SWITCH DEVICE?

When all the PC are connected
in same network then we need
switch between them. But it causes
single broadcast domain.

18) ROUTER?

Router breaks the broadcast. Device
in different networks can be connected
with router b/w them.

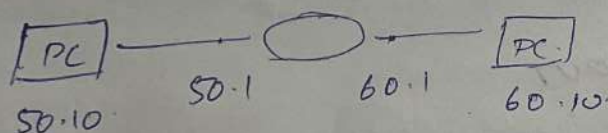
19) SWITCH PACKET FLOW:

When a data enters the switch first it will do address learning $\rightarrow$ noting the receiving port number and source MAC. It will check whether the destination MAC is present in table. If present then it will unicast or else it will broadcast.

20) ROUTER PACKET FLOW:

When a data enters the router, it will check whether there is route for the destination IP in routing table. If route is present it will unicast or else it will broadcast the data.

21) DIRECTLY CONNECTED NETWORKS:



Here 50 & 60 networks are directly connected to the router, so no need to give any static route.

22) TELNET COMMAND

line vty 0 4 $\rightarrow$ devices connected.
transport input telnet
login local
exit

23) CONSOLE PORT AUTHENTICATION:

Username Bugs Password 123
line console 0
login local
exit

24) SYSTEM ROUTE TABLE: 129.0.129.0

Network	Network	Gateway
129.0.0.0	255.255.0.0	129.0.129.0
129.0.129.0	255.255.255.255	129.0.0.1

25) DESTINATION HOST UNREACHABLE

When there is no route for the
DIP in system route table, then this
error occurs.

26) REQUEST TIMED OUT

When data is sent from 'A' PC to
'B' PC, but it cannot reply because
there is no route in system route table for
'B' PC.

27) DEFAULT GATEWAY:

It acts as an entry or exit point
allowing devices on LAN to communicate
with external points in Internet

28) STATIC ROUTE:

Static routes are the manual routes given by user in the system router table or routing table.

29) COMMAND:

ip address	50.0.0.0	255.0.0.0	50.0.0.1
	↓	↓	↓
	N/W	S/M	D/G

30) Loadsharing:

When there are multiple routes for the destination IP in routing table, then it will look for highest matching bits, if all have same matching bits then loadsharing happens.

- per packet
- per destination.

31) COMMANDS: ip load-sharing per-destination
ip load-sharing per-packet.

32) Layer-3 loop: / TTL- importance

When opposite gateways are given the the data cannot exit through default gateways and keep looping. This is L3 looping issue. So, TTL was the solution. Every time routing occurs $TTL \Rightarrow -1$. When $TTL = 0$, Time to live expires.

34) ARP in L2 and L3

ARP $\Rightarrow$ AND function $\Rightarrow$ $\begin{cases} \rightarrow \text{Same N/W (L2)} \\ \rightarrow \text{Different N/W (L3)} \end{cases}$

In L2, no router is involved, so to find destination MAC we can give destination IP of target directly. But in L3 routers are involved so we cannot find DMAC directly. In ARP we should give target IP as default gateway, because ARP is a broadcast & router is unicast.

35) MTU: Maximum Transmission Unit

There is a limited size of data that we can share using network.

36) Segment $\rightarrow$ When we share a data of bigger size, it will break down into segments first and then only it reaches the target.

SYN NO $\rightarrow$ The segment all given sequence nos.

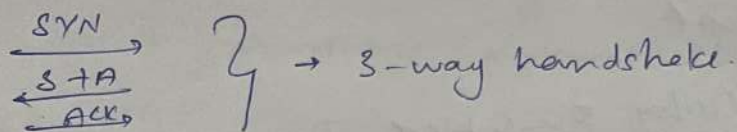
Syn. no is the starting no.

Window size $\rightarrow$ Total no. of packets sent.

(L2)

37) 3-way handshake

It uses TCP protocol. When 2 devices want to share data 3-way handshake is done.



(L3)

38) LINUX & WINDOWS

* Linux uses CLI & it is an open-source.

* Windows uses GUI and it is not open-source it is licensed.

39) KERNEL

It is the important part of OS. It has important programming files. It is like engine of the car.

40) CLIENT OS

The OS which receives the service from server.

SERVER OS

The OS which hosts the service and gives it to client.

41) Web service

Software system that allows 2 different applications or devices to communicate & exchange data with each other over internet.

42) DNS - Domain Name System.

It is a service used to provide name to IP address. It works with UDP protocol.

43) FQDN: Fully Qualified Domain Name

www.hotstar.com

The diagram shows the breakdown of the FQDN 'www.hotstar.com' into its components. A bracket under 'www' points to the label 'Host'. A bracket under 'hotstar' points to the label 'Domain'. A bracket under 'com' points to the label 'Zone'.

44) Packet flow:

labs.surretnetwork.com.

DNS query →

← DNS reply

SYN →

← SYN+ACK

ACK →

45) Port no: A 16-bit virtual identifier ranging from 0 to 65,535 that directs network data to the correct application or service on a device.

46) NTP - 123

IMAP - 143 & 993

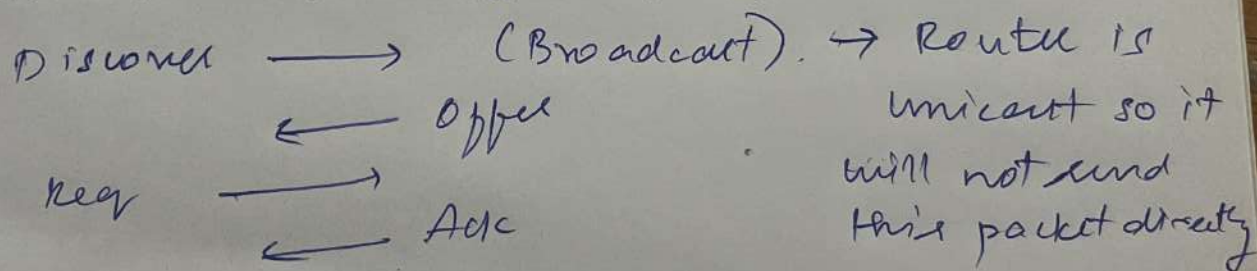
POP3 - 110 & 995

47) DHCP: Dynamic Host Configuration Protocol.

we cannot assign IP configurations for all devices manually, so we can give IP configurations dynamically using DHCP protocol. When we give obtain IP address in client device automatically IP will be assigned using DORA process.

48) DHCP on same & different n/w:

If DHCP is connected on same network then if we give obtain IP directly it will get IP configurations, but in different network router is present between client and server.



So DHCP relay agent is used to convert the broadcast into unicast.

49) Command - DHCP relay agent.

```
interface ethernet 0/0
  ip dhcp helper-address 50.0.0.1
```

DHCP server
↑